

INTEGRATED OPPORTUNITY & LEARNER DASHBOARD

Excelrate – Data Visualization Associate
Week 3 Documentation

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Tools: **PostgreSQL | Google Looker Studio | Google Sheets**

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1. Introduction

The Week 3 project focuses on developing an interactive Business Intelligence (BI) dashboard using integrated learner and opportunity datasets. The primary objective is to transform raw data into meaningful visualizations that support informed decision-making. The project combines data integration, data validation, and dashboard development to provide stakeholders with a centralized view of learner applications, opportunity performance, application trends, and engagement metrics.

The datasets were integrated and validated using PostgreSQL before being imported into Google Looker Studio for visualization. During the data preparation phase, the datasets were examined for missing values, inconsistent data types, duplicate records, and data quality issues. After successful integration, the dashboard was designed using interactive scorecards, charts, and filters to provide a comprehensive overview of the available information.

The final dashboard enables users to monitor important business metrics such as the total number of applications, available opportunities, learner participation, approved applications, application fees, opportunity categories, lead sources, assigned cohorts, and monthly application trends. SQL queries were used throughout the project to validate every KPI displayed in the dashboard, ensuring consistency between the database and the visualization.

This project demonstrates the complete workflow of Business Intelligence development, including data integration, SQL-based validation, dashboard design, and business insight generation.

2. Project Objectives

The primary objective of this project is to develop an interactive dashboard that provides meaningful insights from the integrated learner and opportunity datasets. The dashboard enables stakeholders to analyze learner participation, monitor opportunity performance, and evaluate application trends through visual analytics.

The specific objectives of this project are:

- Integrate learner and opportunity datasets into a single consolidated dataset.
- Clean and validate the data using PostgreSQL before visualization.
- Create an interactive dashboard using Google Looker Studio.
- Design KPI scorecards to summarize important business metrics.
- Develop charts that visualize application trends, opportunity categories, learner status, assigned cohorts, and lead sources.
- Validate dashboard metrics using SQL queries to ensure accuracy.
- Generate observations, business insights, and recommendations based on the dashboard findings.
- Demonstrate a complete Business Intelligence workflow from raw data to interactive reporting.

3. Dataset Overview

The dashboard is based on an integrated dataset created by combining the Learner dataset and the Opportunity dataset. These datasets contain information about learner applications, internship opportunities, application status, assigned cohorts, fee details, and opportunity information.

The integration process was performed using PostgreSQL to establish relationships between the two datasets and create a single dataset suitable for dashboard development.

Dataset Summary

Attribute	Description
Dataset Name	Integrated Data
Database	PostgreSQL
Visualization Tool	Google Looker Studio
Total Application Records	14,799
Unique Opportunities	5,730
Unique Learners	5,730

Important Fields Used

The following fields were used for dashboard development:

Field Name	Purpose
opportunity_id	Unique identifier for each opportunity
opportunity_name	Name of the internship/opportunity
opportunity_category	Category of opportunity
status	Current application status
assigned_cohort	Assigned learner cohort
apply_date	Date of application
fee	Opportunity fee
from_where_did_you_hear_about_us	Lead source
sk	Unique learner identifier

4. Data Integration Process

The data integration process involved combining the learner and opportunity datasets into a unified dataset that could be used for visualization and reporting. PostgreSQL was used throughout the integration process due to its efficient handling of relational data and SQL-based validation.

The integration process consisted of the following steps:

Step 1 – Importing the Datasets

The learner and opportunity datasets were imported into PostgreSQL as separate tables. During this stage, the table structures, column names, and data types were verified.

Step 2 – Data Cleaning

Several preprocessing activities were performed before integration, including:

- Removing duplicate records
- Handling missing values
- Correcting inconsistent data formats
- Converting text fields into appropriate data types
- Verifying primary and reference keys

Step 3 – Dataset Integration

The cleaned learner and opportunity datasets were integrated to create a single dataset named **integrated_data**. The integration combined learner information with opportunity details, allowing comprehensive analysis within a single dashboard.

Step 4 – SQL Validation

After integration, SQL queries were executed to verify:

- Total number of records
- Unique learners
- Unique opportunities
- Application status counts
- Fee totals
- Opportunity categories

This validation ensured that the integrated dataset accurately represented the original data sources.

Step 5 – Export and Visualization

The validated integrated dataset was exported as a CSV file and uploaded to Google Looker Studio. Field data types and aggregations were configured before designing the dashboard visualizations.

5. Dashboard Design

The dashboard was designed in Google Looker Studio to provide an interactive and user-friendly interface for analyzing learner and opportunity data. The design focuses on presenting key performance indicators and business trends through a combination of scorecards, charts, filters, and tables.

The dashboard follows a structured layout, beginning with KPI scorecards that provide an overview of important metrics, followed by visualizations that explain application distribution, learner participation, and opportunity performance.

Dashboard Components

The dashboard contains the following KPI scorecards:

- Total Applications
- Total Opportunities
- Total Learners
- Approved Applications
- Total Fees

The following visualizations were created:

- Top 10 Opportunities by Applications
- Opportunity Category Distribution
- Applications by Status
- Top Assigned Cohorts
- Lead Source Distribution
- Monthly Application Trend

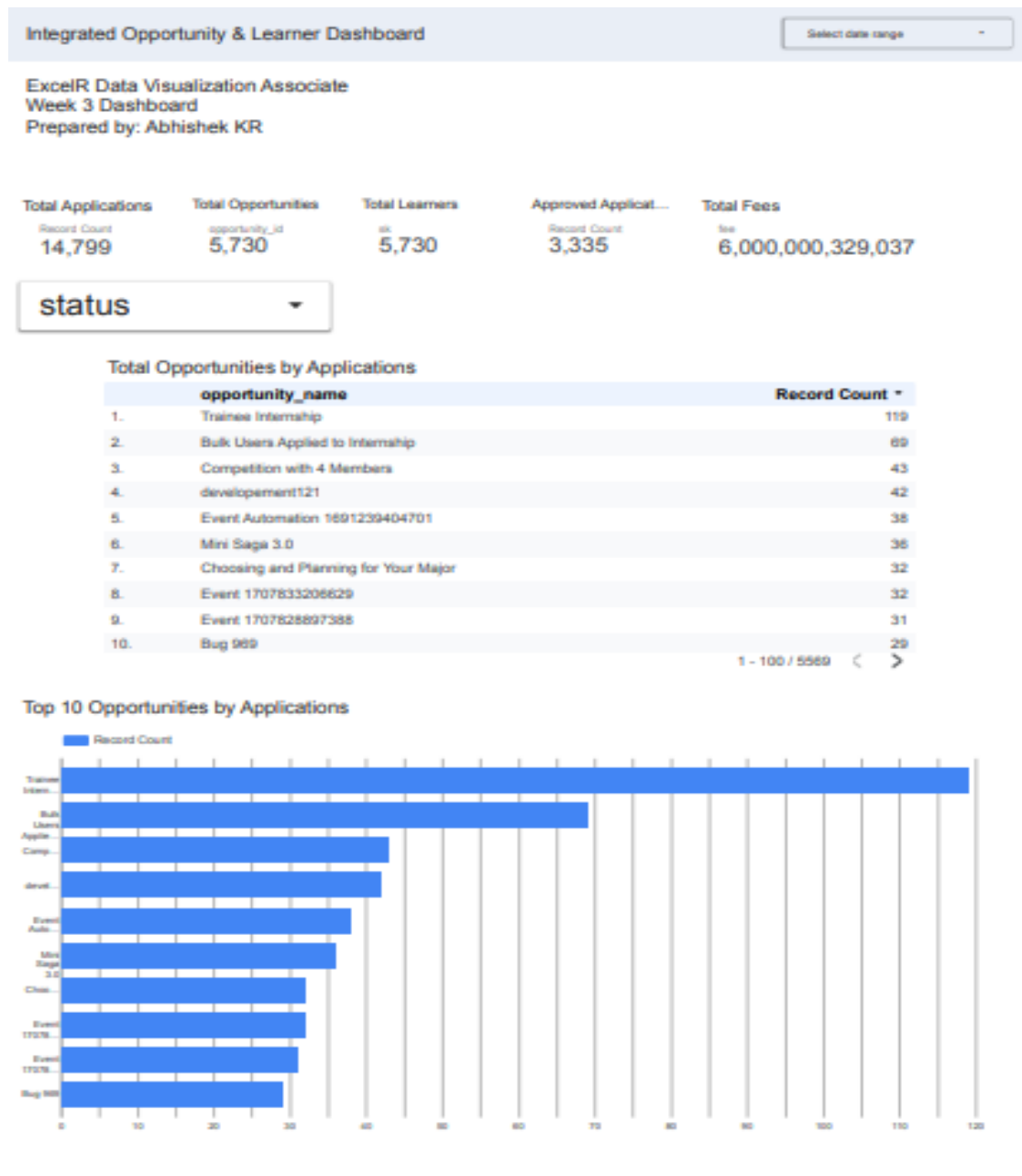
Interactive filters were also included to improve usability:

- Date Range Filter
- Status Filter

The dashboard was designed with consistency in color, layout, and typography to improve readability and user experience. Each visualization was validated using SQL queries to ensure that the values displayed matched the results obtained from PostgreSQL.

The final dashboard provides stakeholders with an efficient tool for monitoring learner engagement, opportunity demand, recruitment performance, and application trends, enabling informed business decisions through interactive data visualization.

Google Looker Studio dashboard screenshot here.



6. Dashboard Components

The dashboard was developed in Google Looker Studio using the integrated dataset created from the learner and opportunity tables. The primary objective of the dashboard is to present key business metrics in an interactive and visually meaningful format. Each dashboard component was designed to answer specific business questions and was validated using SQL queries executed in PostgreSQL.

Dashboard KPI Scorecards

The dashboard contains the following Key Performance Indicators (KPIs):

KPI	Purpose
Total Applications	Displays the total number of application records.
Total Opportunities	Displays the number of unique opportunities available.
Total Learners	Displays the number of unique learners.
Approved Applications	Shows the total approved applications based on status code 1070.
Total Fees	Displays the total fee amount collected from all applications.

Dashboard Visualizations

The dashboard includes the following charts:

Visualization	Purpose
Top 10 Opportunities	Identifies opportunities receiving the highest number of applications.
Opportunity Category Distribution	Displays application distribution across opportunity categories.
Applications by Status	Shows the number of applications in each workflow status.
Top Assigned Cohorts	Identifies cohorts with the highest learner participation.
Lead Source Distribution	Displays how learners discovered the opportunities.
Application Trend	Shows monthly application trends over time.

7. SQL Validation Queries

To ensure the accuracy of the dashboard, every KPI and visualization was validated using SQL queries in PostgreSQL. The SQL results were compared directly with the values displayed in Google Looker Studio.

7.1 Total Applications

SQL Query

```
SELECT COUNT(*) AS total_applications  
FROM integrated_data;
```

Expected Result

	total_applications bigint
1	14799

14,799 Applications

7.2 Total Opportunities

SQL Query

```
SELECT COUNT(DISTINCT opportunity_id) AS total_opportunities  
FROM integrated_data;
```

Expected Result

	total_opportunities bigint
1	5730

5,730 Opportunities

7.3 Total Learners

SQL Query

```
SELECT COUNT(DISTINCT sk) AS total_learners  
FROM integrated_data;
```

Expected Result

	total_learners bigint
1	5730

5,730 Learners

7.4 Approved Applications

SQL Query

```
SELECT COUNT(*) AS approved_applications  
FROM integrated_data  
WHERE status = '1070';
```

Expected Result

	approved_applications bigint
1	3335

3,335 Approved Applications

7.5 Total Fees

SQL Query

```
SELECT SUM(fee::numeric) AS total_fees  
FROM integrated_data  
WHERE fee IS NOT NULL;
```

Expected Result

	total_fees numeric
1	6000000329037

6,000,000,329,037

7.6 Top 10 Opportunities

SQL Query

```
SELECT  
opportunity_name,  
COUNT(*) AS applications  
FROM integrated_data  
GROUP BY opportunity_name  
ORDER BY applications DESC  
LIMIT 10;
```

Expected Result

	opportunity_name text	applications bigint
1	Trainee Internship	119
2	Bulk Users Applied to Internship	69
3	Competition with 4 Members	43
4	developement121	42
5	Event Automation 1691239404701	38
6	Mini Saga 3.0	36
7	Choosing and Planning for Your Ma...	32
8	Event 1707833206629	32
9	Event 1707828897388	31
10	Bug 969	29

7.7 Opportunity Category Distribution

SQL Query

```
SELECT  
  
opportunity_category,  
  
COUNT(*) AS applications  
  
FROM integrated_data  
  
GROUP BY opportunity_category  
  
ORDER BY applications DESC;
```

Expected Result

	opportunity_category text	applications bigint
1	Internship	5180
2	Competition	2202
3	Career	2053
4	Event	1747
5	Course	1322
6	Engagement	944
7	Masterclass	910
8	JobSimulation	423
9	Xploreu	7
10	Program	7
11	NULL	4

7.8 Applications by Status

SQL Query

SELECT

status,

COUNT(*) AS applications

FROM integrated_data

GROUP BY status

ORDER BY applications DESC;

Expected Result

	status text	applications bigint
1	[null]	5749
2	1070	3335
3	1010	1791
4	1120	1123
5	1030	694
6	1020	597
7	1110	435
8	1055	348
9	1050	326
10	1040	196
11	1011	101
12	1080	94
13	1100	4
14	1090	2
15	103	2
16	1060	1
17	110	1

7.9 Top Assigned Cohorts

SQL Query

SELECT

assigned_cohort,

COUNT(*) AS learners

FROM integrated_data

GROUP BY assigned_cohort

ORDER BY learners DESC

LIMIT 10;

Expected Result

	assigned_cohort text	learners bigint
1	[null]	9247
2	B9ZYF9D	68
3	B28EPFO	62
4	B360ZU8	32
5	BAD40TQ	31
6	BC5AWSN	30
7	B202074	22
8	B905078	20
9	BSQNZSD	19
10	BMSFPMW	18

7.10 Lead Source Distribution

SQL Query

```
SELECT
```

```
from_where_did_you_hear_about_us,
```

```
COUNT(*) AS learners
```

```
FROM integrated_data
```

```
GROUP BY from_where_did_you_hear_about_us
```

```
ORDER BY learners DESC;
```

Expected Result

	from_where_did_you_hear_about_us text	learners bigint
1	[null]	5886
2	Social Media	3282
3	Illinois Institute of Technology	2034
4	GlobalShala Website	845
5	Saint Louis University	688
6	Google Search	589
7	social Media	533
8	My Friend Or Relative	309
9	Illinois Tech	203
10	My Friend/Relative	192
11	Northeastern University	128
12	Global Enrichment Program	46
13	Others	38
14	Other	16
15	Rochester Institute of Technology	8
16	Hiring Platforms	1
17	SLU	1

7.11 Monthly Application Trend

SQL Query

SELECT

DATE(DATE_TRUNC('month', apply_date::timestamp)) AS
application_month,

COUNT(*) AS applications

FROM integrated_data

WHERE apply_date IS NOT NULL

GROUP BY DATE(DATE_TRUNC('month', apply_date::timestamp))

ORDER BY DATE(DATE_TRUNC('month', apply_date::timestamp));

Expected Result

	application_month date	applications bigint
1	2022-06-01	4
2	2022-07-01	4
3	2022-08-01	17
4	2022-09-01	8
5	2022-10-01	31
6	2022-11-01	12
7	2023-01-01	84
8	2023-02-01	153
9	2023-03-01	216
10	2023-04-01	308
11	2023-05-01	248
12	2023-06-01	544
13	2023-07-01	295
14	2023-08-01	678
15	2023-09-01	238
16	2023-10-01	343
17	2023-11-01	335
18	2023-12-01	194
19	2024-01-01	193
20	2024-02-01	267
Total rows: 46		Query complete 00:00

8. SQL Query Explanations

Each SQL query was designed to validate a specific KPI or visualization shown in the dashboard.

Total Applications

The COUNT(*) function counts every application record available in the integrated dataset.

Total Opportunities

COUNT(DISTINCT opportunity_id) ensures that each opportunity is counted only once, eliminating duplicate application records.

Total Learners

COUNT(DISTINCT sk) calculates the number of unique learners who submitted applications.

Approved Applications

The WHERE status='1070' condition filters only approved applications before counting the records.

Total Fees

The SUM() function calculates the total fee collected. Since the fee field was stored as text, it was converted to numeric using fee::numeric.

Top Opportunities

The dataset is grouped by opportunity name, and the number of applications for each opportunity is counted and sorted in descending order.

Opportunity Categories

Applications are grouped according to opportunity category to understand category popularity.

Application Status

Grouping by status provides insight into the application workflow distribution.

Assigned Cohorts

Applications are grouped based on assigned learner cohorts to identify the most active cohorts.

Lead Sources

Applications are grouped according to the learner's source of information to evaluate recruitment effectiveness.

Application Trend

Application dates are converted into monthly values using `DATE_TRUNC()` to visualize monthly trends.

9. Validation Tables

The SQL outputs were compared with the dashboard metrics to verify accuracy.

KPI Validation

KPI	SQL Result	Dashboard Result	Validation
Total Applications	14,799	14,799	☐ Match
Total Opportunities	5,730	5,730	☐ Match
Total Learners	5,730	5,730	☐ Match
Approved Applications	3,335	3,335	☐ Match
Total Fees	6,000,000,329,037	6,000,000,329,037	☐ Match

Chart Validation

Dashboard Chart	SQL Validation	Status
Top 10 Opportunities	Verified using GROUP BY opportunity_name	☐
Opportunity Category Distribution	Verified using GROUP BY opportunity_category	☐
Applications by Status	Verified using GROUP BY status	☐
Top Assigned Cohorts	Verified using GROUP BY assigned_cohort	☐
Lead Source Distribution	Verified using GROUP BY from_where_did_you_hear_about_us	☐

Dashboard Chart	SQL Validation	Status
Application Trend	Verified using DATE_TRUNC() monthly aggregation	☐

10. KPI Verification

The KPI verification process confirmed that the dashboard accurately represents the underlying integrated dataset. Every KPI displayed in Google Looker Studio was validated against PostgreSQL using SQL queries. The comparison demonstrated complete consistency between the database results and the dashboard values, confirming the correctness of the dashboard configuration and aggregation methods.

This validation process increased confidence in the reliability of the dashboard and ensured that stakeholders can use it for decision-making without concerns regarding data accuracy. The use of SQL verification also helped identify and resolve earlier issues related to duplicate counts, text-based numeric fields, and aggregation settings, resulting in a fully validated and accurate business intelligence dashboard.

11. Chart-by-Chart Analysis

This section explains each visualization included in the Google Looker Studio dashboard. Each chart was designed to answer a specific business question and provide meaningful insights into learner applications, opportunity performance, recruitment effectiveness, and application trends.

11.1 Total Applications (Scorecard)

Purpose

The **Total Applications** scorecard displays the total number of application records available in the integrated dataset.

Business Importance

This KPI provides an overall measure of platform activity and serves as the primary indicator of learner engagement.

Dashboard Result

14,799 Applications

Analysis

The scorecard indicates that a total of **14,799** application records were processed in the integrated dataset. This value represents all learner applications submitted for various opportunities.

11.2 Total Opportunities (Scorecard)

Purpose

Displays the number of unique opportunities available in the integrated dataset.

Dashboard Result

5,730 Opportunities

Analysis

This KPI measures the number of distinct internship, competition, career, and event opportunities available for learners.

11.3 Total Learners (Scorecard)

Purpose

Displays the total number of unique learners who applied.

Dashboard Result

5,730 Learners

Analysis

The scorecard helps evaluate learner participation and platform reach by counting unique learner records.

11.4 Approved Applications (Scorecard)

Purpose

Displays the number of approved applications (Status Code **1070**).

Dashboard Result

3,335 Approved Applications

Analysis

Approved applications represent learners who successfully progressed through the application process.

11.5 Total Fees (Scorecard)

Purpose

Displays the cumulative fee value recorded in the integrated dataset.

Dashboard Result

6,000,000,329,037

Analysis

This KPI provides a financial overview of the application data and represents the aggregated fee values after converting the field to a numeric data type.

11.6 Top 10 Opportunities by Applications

Purpose

This visualization ranks opportunities according to the number of learner applications received.

Business Importance

It helps identify the most popular opportunities and learner preferences.

Analysis

The visualization highlights the opportunities that consistently attract the highest application volume, enabling administrators to recognize successful programs and improve less popular opportunities.

11.7 Opportunity Category Distribution

Purpose

Shows the distribution of applications across different opportunity categories.

Business Importance

Understanding category popularity assists organizations in planning future programs.

Analysis

Internship opportunities represent the largest share of learner applications, followed by Career and Competition categories. This indicates that learners primarily seek career-oriented opportunities.

11.8 Applications by Status

Purpose

Displays the distribution of applications across different workflow status codes.

Business Importance

The chart enables administrators to monitor application progress and workflow efficiency.

Analysis

Most applications belong to a limited number of workflow stages, demonstrating that learner applications follow a structured approval process.

11.9 Top Assigned Cohorts

Purpose

Displays the cohorts contributing the highest number of learner applications.

Business Importance

The visualization identifies the most active learner groups.

Analysis

Some cohorts contribute significantly more applications than others, indicating differences in learner engagement and participation.

11.10 Lead Source Distribution

Purpose

Shows how learners discovered the opportunities.

Business Importance

Lead source analysis evaluates the effectiveness of recruitment and marketing strategies.

Analysis

Digital channels, institutional referrals, and search platforms contribute the majority of learner applications.

11.11 Monthly Application Trend

Purpose

Displays application growth over time.

Business Importance

Trend analysis assists stakeholders in understanding seasonal demand and planning future recruitment campaigns.

Analysis

The application trend demonstrates fluctuations across different months while showing periods of increased learner participation.

12. Observations

The following observations were derived from the dashboard analysis.

Observation 1

The integrated dataset contains **14,799** learner application records.

Observation 2

A total of **5,730** unique opportunities are available.

Observation 3

The number of unique learners is **5,730**, indicating broad learner participation.

Observation 4

More than **3,300** applications have reached the approved stage.

Observation 5

Internship opportunities receive considerably more applications than other opportunity categories.

Observation 6

Certain learner cohorts contribute a significantly higher number of applications.

Observation 7

Lead source analysis indicates that online platforms and institutional referrals play a major role in attracting learners.

Observation 8

Application activity varies throughout the year, suggesting seasonal enrollment behavior.

13. Insights

The dashboard provides several valuable business insights.

- Learners show a strong preference for internship opportunities over other opportunity categories.
 - The application approval process successfully progresses a substantial number of learners.
 - Marketing channels significantly influence learner participation.
 - Cohort engagement varies considerably, indicating opportunities for targeted learner support.
 - Historical application trends can be used to forecast future demand.
 - SQL validation confirmed that all dashboard metrics accurately represent the integrated dataset.
-

14. Recommendations

Based on the dashboard findings, the following recommendations are proposed.

Recommendation 1

Increase the number of internship opportunities to meet learner demand.

Recommendation 2

Analyze the characteristics of the highest-performing opportunities and replicate successful practices.

Recommendation 3

Improve engagement strategies for cohorts with lower participation.

Recommendation 4

Continue investing in the most effective lead-generation channels.

Recommendation 5

Schedule future recruitment campaigns before historically high application periods.

Recommendation 6

Maintain regular SQL validation to ensure dashboard accuracy after every dataset refresh.

Recommendation 7

Introduce automated dashboard refresh mechanisms to reduce manual updates.

15. Conclusion

The Week 3 project successfully demonstrates the complete Business Intelligence workflow, beginning with data integration in PostgreSQL and ending with the creation of an interactive dashboard in Google Looker Studio. The integrated dataset was cleaned, validated, and transformed into meaningful visualizations that provide valuable insights into learner participation, opportunity performance, application workflow, and recruitment effectiveness.

SQL validation confirmed that every KPI and chart displayed in the dashboard accurately matched the underlying database. This validation process increased confidence in the reliability of the dashboard and ensured that stakeholders can use it for informed decision-making.

Overall, the dashboard serves as an effective reporting solution that enables administrators and decision-makers to monitor key performance indicators, identify trends, evaluate recruitment strategies, and make data-driven business decisions.

16. References

The following tools and resources were used during the completion of this project.

Resource	Purpose
PostgreSQL	Data integration, cleaning, and SQL validation
pgAdmin	Database management and SQL execution
Google Looker Studio	Interactive dashboard development
Google Sheets	Data source management
Microsoft Excel	Data verification and CSV preparation

Resource	Purpose
Microsoft Word	Documentation preparation

17. Project Links

Google Looker Studio Dashboard

Dashboard Link:

<https://datastudio.google.com/reporting/f8f1609d-b52d-4137-af4e-d8f388db7d4d>

Google Sheets Data Source

Google Sheets Link:

<https://drive.google.com/drive/folders/1OjSVLZ2Y9dgtlYYn4DwdqonTnBJB8zpu?usp=sharing>
